An ML-Based Authentication for Privacy Preservation in a Distributed Edge-Enabled Internet of Vehicles

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***Abstract*—In the Internet of Vehicles (IoV), privacy preservation is a major challenge due to the mobility of vehicles and their resource-constrained nature. The limited resources of on-board units (OBUs) and embedded sensors lure the adversaries to launch various types of attacks. Thus, lightweight but reliable authentication schemes need to be designed to combat these attacks. Another major challenge is the scarcity of available bandwidth and excessive delay experienced by vehicles while they communicate with the servers located at the cloud. These servers execute various machine learning (ML) and deep learning (DL) algorithms to extract useful features from upstream traffic of IoV. To addresses these challenges, we propose an ML-based authentication scheme that trains and classifies the vehicles at the edge servers in a distributed manner, preserves the privacy of communicating entities and minimizes the bandwidth consumption and delay experienced by the vehicles. The ML-based approach extends the decision power of vehicles and edge servers to identify adversaries. Our scheme requires that each vehicle participates in an offline phase, where a trusted authority shares a list of MaskIDs and secret keys of legitimate vehicles and edge servers. A timestamp is embedded in the payload of each encrypted message to prune the proposed scheme against well-known adversarial attacks. The simulation results verify the exceptional performance of our scheme in terms of computational overhead, communication overhead, and storage overhead.**

***Index Terms*—Authentication, distributed edge intelligence (DEI), Internet of Vehicles (IoV), machine learning (ML), privacy preservation.**

I. Introduction

INTERNET of Vehicles (IoV) has transformed cities around the globe by providing real-time communication among vehicles, parking lots, pedestrians, and connected infrastructures [1]. In these networks, vehicles equipped with embedded sensors, actuators and on-board units (OBUs) collect useful data and communicate it with the closest roadside units (RSUs) or server modules. The operational capabilities of these vehicles are further augmented by artificial intelligence, particularly machine learning (ML) and deep learning (DL), which analyze and interpret data in real-time [2]. Despite the popularity of ML and DL algorithms, vast amount of data generated by the vehicles are processed by cloud servers. Most of the applications of IoV are delay-sensitive and time-critical that require prompt responses and ample amount of bandwidth. These cloud servers, even with the use of ML and DL algorithms, are unable to provide swift responses to the vehicles that can lead to catastrophic circumstances at the roads. To resolve this challenge, distributed edge intelligence (DEI) has gained significant attention in recent years that migrates the ML and DL algorithms to the edge computing platforms [3]. DEI allows quicker data processing and has lower response time with requests from vehicles handled with minimal delay.

Another major challenge is the security and privacy preservation of data sources, i.e., embedded sensors and OBUs, as well as destination modules, i.e., RSUs and edge servers. The embedded sensors, OBUs and RSUs are resource-constrained and are unable to support computationally complex security and privacy preservation schemes [4]. It would require ample of resources for these devices to securely communicate with the cloud servers. For a resilient ITS framework, an end-to-end communication paradigm is required that ensures the security and privacy of vehicles, OBUs, RSUs, edge and cloud servers, and other peripheral components in real time [5]. ML-based solutions have the ability to address the security and privacy concerns by preserving the integrity of communicating entities with continuous learning [6], [7]. ML and DL-based solutions of DEI ensure the integrity of vehicle-to-edge (V2E) and edge-to-cloud (E2C) communication links. These solutions offer resilient authentication for the interacting devices that preserve the privacy of exchanged data in an end-to-end framework within an IoV ecosystem. Furthermore, these solutions allow for lightweight authentication schemes to be deployed at the resource-constrained devices of IoV as they are no longer required to communicate directly with the cloud servers for the exchange and validation of security credentials.

Various ML and DL-based authentication approaches have been designed in the literature to resolve this challenging issue of privacy preservation within the DEI-enabled IoV paradigm. A DL-based framework was proposed that classified

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the vehicles into legitimate and adversarial vehicles to enhance the operational capacities of the RSUs [8]. This framework enables the RSUs to process only those requests that belong to the legitimate vehicles in an end-to-end paradigm. Likewise, a two-layer-based authentication approach was proposed to conduct real-time security for an IoV infrastructure [9]. Vehicles were divided into two classes: 1) vehicles inside the fog layer that were managed via a sophisticated authentication approach and 2) vehicles outside the fog layer that were monitored via a DL-based scheme. DL has been integrated with the authentication scheme to form a hybrid and smart architecture that can guarantee intrusion detection in the early stages and streamline the operations of vehicles in IoV [10]. Similarly, a hybrid authentication model comprising of ML and feature engineering was designed to make IoV prunes against well-known Denial-of-Service (DoS) attacks by using a cross-layer intrusion detection dataset as a benchmark [7]. The proposed model consists of two phases: 1) first phase focused on the enrichment of the database and feature engineering for analysis and 2) an ML-based phase to detect DoS attacks in a realistic environment of IoV. A blockchain-based authentication scheme was proposed for the artificial intelligent-oriented IoV to realize the advancement of smart cities [11]. Even though the verification of vehicles or edge servers in the IoV has been successfully handled by current authentication schemes, either the privacy of vehicles is jeopardised or the schemes are tailored for particular infrastructures. Thus, an infrastructure-free and privacy-preserving authentication approach was proposed by the authors, equally implementable in different application domains, such as IoV and other networking infrastructures.

In this article, an ML-based authentication scheme is proposed to preserve the privacy of interacting devices of IoV. Given the resource-constrained nature of embedded sensors, OBUs and RSUs, our proposed scheme ensures the privacy of interacting devices while maintaining an extremely low computational, communication and storage overhead. For the authenticity of vehicular devices and edge servers, MaskIDs and secret shared key are used to form a hybrid model. Moreover, the use of timestamps makes the proposed scheme resilient and robust against well-known adversarial attacks. Importantly, vehicular devices and edge servers randomly select MaskIDs from a pool of legitimate MaskIDs, shared in an offline phase. Hence, the privacy of interacting devices is preserved while sharing sensitive information with the edge server or trusted authority (TA) in the cloud. The main contributions of this article are as follow.

1) An ML-based authentication approach for the classification of adversaries from the legitimate devices, i.e., vehicles and edge servers. The classification is based on three metrics, which are average transmission time, MaskIDs and a secret key.
2) A 64-bit data encryption standard (DES) algorithm, which requires only two inputs for encryption and decryption: a) a 64-bit MaskID and b) a 48-bit secret key. The use of lightweight crypto-primitives means that the scheme is extremely lightweight with low overhead in terms of computation, communication and storage.
3) Random selection of MaskIDs, preferably from a pool of available legitimate MaskIDs, to preserve the privacy of the source and destination. A timestamp is embedded to enhance the security of the proposed scheme against well-known adversarial attacks.

The remainder of this article is arranged as follows. A comprehensive literature review of the most relevant contents from the literature is presented in Section II. Our proposed ML-based authentication approach, its working principles, and its detailed description is provided in Section III, which is followed by a sophisticated security analysis in Section IV. In Section V, comprehensive experimental results along with resilience of our work against various attacks is presented. Finally, the concluding remarks are presented in Section VI.

II. Literature Review

A message authentication code (MAC) was designed by [12], where the authenticity of communicating devices was ensured with minimal possible processing and communication overheads in IoV. In this approach, hard and soft keys are updated frequently to prune the proposed scheme against data leakage problem that occurs due to the side channel attacks. A conditional authentication scheme that ensures the privacy of communicating entities is presented in [13], where the devices are required to carry out the legitimacy of neighboring modules, i.e., vehicles and RSUs. This scheme enables communicating entities to be pruned against active and passive attacks launched by the adversary devices in the IoV. A TA-free vehicle-to-RSU authentication scheme, where the vehicles and RSUs are required to be agreed upon a secured key for encryption and decryption, was proposed in [14]. Due to the dense deployment of vehicles in IoV, the secret key may probably be leaked out that can compromise the whole network. To resolve this challenge, a noncryptographic authentication scheme, which is based on the enhanced DL framework, was designed to ensure the authenticity of member devices in IoV [15]. A DL-based framework for vehicles' classification into legitimate and misbehaving classes was developed to enhance the operational capabilities of the RSUs in IoV [8]. This approach has enabled RSUs to process only those requests belonging to the legitimate class. Likewise, a two-layered identity authentication approach was designed to conduct real-time security analysis in IoV. Vehicles were divided into two classes, i.e., 1) vehicles inside the Fog and managed through a sophisticated security authentication layer and 2) vehicles outside the Fog and monitored through a DL-based mechanism [9]. DL has been integrated with the authentication scheme to form a hybrid and smart architecture, which can guarantee intrusion detection in the earliest stages and streamline the operations of IoV [10].

A supervised ML-based model was proposed by authors in [7] to enhance the misbehavior detection capabilities of the embedded devices in IoV. Second, this approach was based on the data-centric approach rather than node-centric, as the latter becomes ill-fit in different scenarios. RSU placement is a crucial task and requires extra attention, therefore, an ML-based approach was presented by [16] to resolve this

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issue. Moreover, a malicious IoV detection algorithm was integrated to prune the IoV against well-known intruder attacks. Similarly, a transfer learning-enabled intrusion detection scheme was developed for the IoV [17]. It has two submodels: 1) a cloud-assisted model that ensures IoV-enabled cloud has the potential to provide the required data and 2) a local update model that is applicable in scenarios, where the IoV-enabled cloud is unable to provide the required data. A hierarchical self-encoder model, based on a three-layered neural network and support vector machine, was proposed to ensure fusion security in IoV [18]. Likewise, a smart, secured, and intelligent framework for the IoV was designed that is primarily based on graph convolutional networks. In this networking infrastructure, communication among vehicles and RSUs was secured via a blockchain-based approach [19].

All these existing schemes have their own shortcomings. First, most of them are designed for a specialized framework and are not suitable in other application scenarios. Second, these schemes either ensure the authenticity of communicating entities or their privacy but not both at the same time. Importantly, many of these schemes are not secured against a comprehensive list of vulnerable attacks. Third, these schemes are excessively complex in terms of computational, communication and storage overheads and mostly do not focus on enhancement of QoS during the transmission of data.

III. Proposed Scheme for Secured and Distributed Edge Intelligence in IoV

In IoV, vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication is carried out over a shared transmission medium that is highly vulnerable to numerous adversarial attacks. Given the limited processing capabilities of OBUs and embedded sensors, authentication is a challenging task in V2V and V2I. Moreover, the mobility of these vehicles adds another layer of complexity to authentication. Our proposed architecture for privacy-preserving authentication uses encrypted messages to safeguard communication over a shared transmission medium as shown in Fig. 1. The vehicles are classified into two categories: 1) legitimate and 2) intruders (adversaries), using a sophisticated ML-based classification. Through continuous interaction with RSUs, the edger servers classify the vehicles and control their mobility. These edge servers constitute our mobile edge computing framework (MEC). Our proposed authentication approach has two phases along with a prerequisite offline phase. These phases are ML-based adversary detection and online verification. A detailed description along with working principles of these phases is discussed in this section. Please note that this article does not cover V2V authentication as it is out of the scope of this article and there exist a wealth of research in this domain.

A. Offline Phase

The offline phase is based on the realistic assumption that an adversary or set of adversaries do not have access to the OBUs and embedded sensors of vehicles as communication is carried out over a secured transmission channel, particularly in a private network. Moreover, communication in this phase

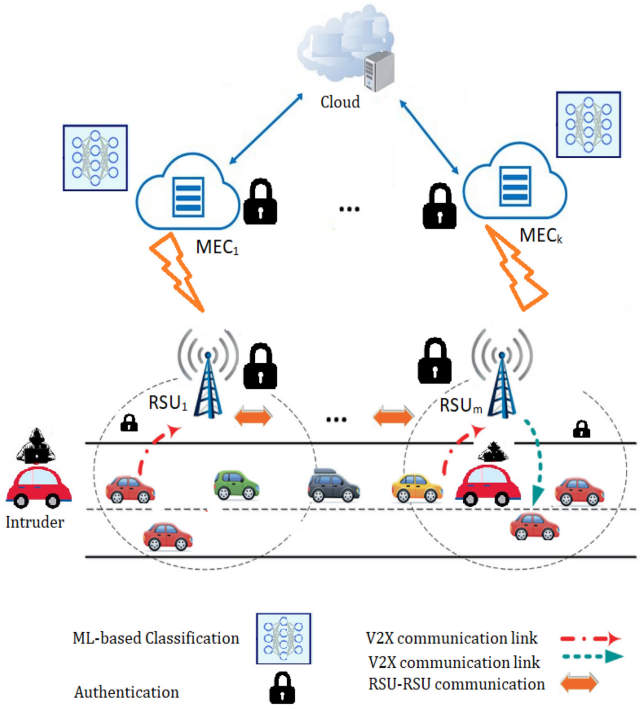


Fig. 1. Architecture of a secured DEI for IoV.

is carried out using plaintext and does not require encryption standards. In this phase, each vehicle (V_i) and edge server (S_j) generate a 64-bit ID through a substitution-box (S-box) by using their MAC addresses and share it with the TA via an updated message, which is generated and broadcast by the TA as a plaintext. The nonlinearity of the S-box has a strong correlation to the security of our scheme. In other words, a stronger nonlinearity means that a resilient security measure is in place. It is computed using the boolean function f, which is defined as the approximate distance, i.e., hamming distance between the affine functions. An affine function is defined as a linear function along with a constant, whereas its graph is a straight line. Next, the Walsh–Hadamard transformation (WHT) [20] is used to find the correlation ratio between the boolean and linear functions. Nonlinearity is calculated using

$$NL = \frac{1}{2}\left(2^n - \mathrm{WHT}(\max(f))\right) \tag{1}$$

where NL represents the overall nonlinearity of the respective device, be it an RSU or vehicle. In our scheme, TA generates a MaskID for each V_i and S_j and stores them using two separate lists, one for V_i and the second for S_j, respectively. These lists along with the secret key of 64-bit (λ) are shared with each V_i as they are crucial for the authentication and verification of V_i and S_j. A graphical representation of this phase is shown in Fig. 2.

B. Machine Learning-Based Adversary Detection

In our scheme, each S_j and V_i (via RSU) is equipped with an ML algorithm to classify adversaries from legitimate ones. For this purpose, these devices are trained on actual data, preferably embedded in the payload of every request message,

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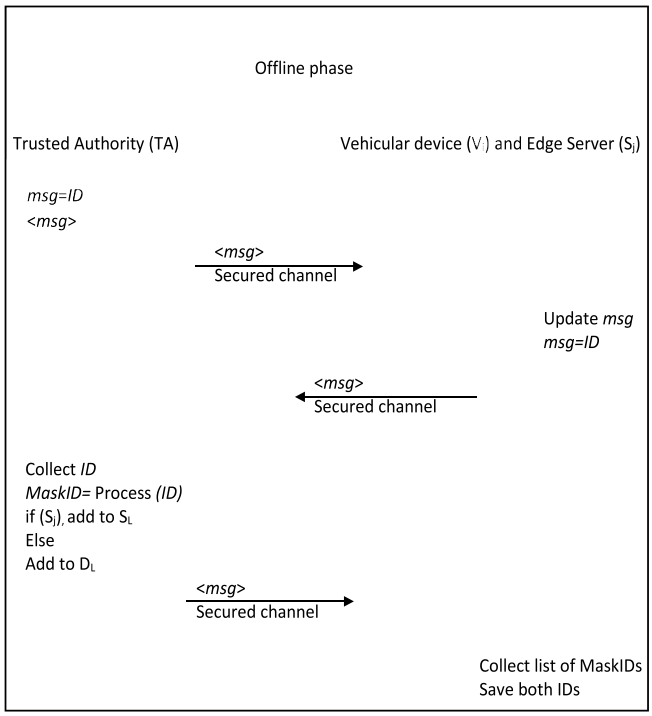


Fig. 2. Offline phase of our secured DEI in IoV.

in the offline phase. Initially, the dataset is required to be processed by a well-known feature extraction algorithm to separate the dependent and independent variables of a real-time dataset. In the proposed scheme, the independent variables are λ, *MaskID*, and $\triangle T$, whereas the requesting device, i.e., a legitimate vehicle or an adversarial vehicle, is the dependent variable. Additionally, another important aspect is the missing values in a particular dataset, which is quite common in IoV where the expected probability of packet loss ratio is extremely high. Therefore, the dataset must be tuned before initiating the actual processing, i.e., classification of requesting vehicles in the IoV. For this purpose, the entire chunk of data related to the missing values may be discarded, however, it is not realistic in the real-time applications of IoV. Another approach is to compute the mean of a particular column, especially those containing missing values. This approach ensures that missing values are replaced by the mean value. Furthermore, in classification, classifiers in a dataset has a strong correlation to the labels as represented by

$$\text{Classifiers} = N \times \frac{(N-1)}{2} \tag{2}$$

where N represents the number of labels in a given dataset. This classification is based on three valuable metrics, i.e., 1) average transmission time $\triangle T = T_R$-T_G, where T_G and T_R represent the time interval of message generation and message reception; 2) MaskID in the payload of a message that belongs to the registered IDs, i.e., $MaskID \in ServerReg(MaskID)$ or $MaskID \in DeviceReg(MaskID)$; and 3) decryption of messages with λ, where $\lambda(msg)$ results in a plaintext message. Next, these metrics are assigned weights, i.e., $\triangle T = x$, $\lambda = y$, and $MaskID = x$, where $x = 0.3$ and $y = 0.4$, and this decision of assignment is based on expected value of the expression in

$$\text{Auth}(V_i) = \frac{(\triangle T \times x) + (\lambda(msg) \times x) + (\text{MaskID} \times x)}{3 \times (2x + y)}. \tag{3}$$

A requesting vehicle V_i is authentic if its minimum score, based on (3), is greater than or equal to the defined threshold value of δ. Thus, each S_j and V_i is trained on the training dataset, which is large enough and consists of both legitimate and fraudulent request messages. In the testing phase, each V_i and S_j is passed through a rigorous testing procedure containing legitimate and fraudulent requests preferably with different ratios. In the testing phase, each V_i and S_j is thoroughly examined by passing messages generated by both legitimate and adversary vehicles. For this purpose, initially, multiple request messages are generated, which consist of both legitimate and fraudulent ones and shared with the concerned S_j that uses an ML-based authentication to classify these messages into legitimate and fraudulent ones, which is based on the approximate value of (3). Furthermore, V_i are tested by sending response messages from the legitimate S_j and an adversary that pretends as a legitimate S_j in the IoV. If V_i and S_j pass this testing phase with an approximately higher accuracy ratio, then the proposed methodology designed for V_i and S_j is assumed to be reliable. Moreover, the proposed model is also checked in scenarios where every requesting vehicle is an adversary and not a single message is generated by a legitimate vehicle V_i. The accuracy of our ML-based model, i.e., classification, is computed using

$$\text{Accuracy} = \frac{T_P + T_F}{T_O}. \tag{4}$$

Here, T_P and T_F represent the true positive and true negative, whereas T_O represents the total number of observations. Moreover, the precision ratio is computed using

$$\text{Precision} = \frac{T_P}{T_P + T_F}. \tag{5}$$

After the training and testing phases, S_j and V_i can easily distinguish legitimate requests from fraudulent ones with maximum accuracy and precision.

C. Verification Phase

Each V_i and S_j pass through a sophisticated verification process where their legitimacy is checked. For this purpose, various verification metrics are utilized that include but are not limited to IDs, MaskIDs, MAC addresses, and one-time passwords (OTPs). Furthermore, this process needs to be completed prior to the actual communication session, where the real-time data are shared among V_i and S_j. As discussed in Section III-A, each vehicle and edge server receives a list of all verified V_i along with all verified S_j. These lists are crucial in scenarios where the authenticity of the requested module, either an V_i or S_j, needs to be verified. Moreover, these lists allow the differentiation of legitimate V_i from adversaries that are eager to steal valuable credentials or disturb the usual communication sessions in DEI. Thus, if an authentic V_i is interested in establishing a secured communication session with a legitimate S_j, it selects a MaskID in a random fashion

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from the authentic device list (D_L), as shown in (6). The MaskIDs are shared in the offline phase and appended to the payload of an encrypted message

$$\text{MaskID} = \text{rand}(D_L). \tag{6}$$

V_i and S_j are not bound to share their actual MaskIDs as the encrypted messages are read by other trusted devices deployed in their vicinity. Additionally, V_i and S_j are interested to preserve their privacy while providing the required services in an effective manner. The random selection of MaskIDs ensures the privacy of the requesting devices, which is a requirement imposed by various applications. Because these messages are transmitted over a shared medium that is highly susceptible to intruder attacks, a sophisticated encryption scheme is used for this purpose. Our encryption scheme converts the plaintext messages of V_i and S_j into ciphertexts to make them unreadable for intruders. For this purpose, we have adopted a 64-bit DES, that is not only secure but it is extremely reliable as well, especially for the resource-constrained devices of DEI. The DES algorithm needs two inputs in order to encrypt a particular plaintext message: 1) a 64-bit plaintext (in this case, *MaskID*) and 2) a 48-bit λ.

Initially, plaintext passes through the initial permutation process (IP), where bits are rearranged according to the given permutation by producing a permuted text as an output. Next, the permuted text is split into two equivalent parts: 1) L (left) and 2) R (right). Because the length of the subkey K_i is 48-bit, a 32-bit R is required to be expanded to 48-bit, which is carried out using an expansion permutation box (P-box). Finally, an XOR operation is carried out using K_i and expanded R as shown in

$$E_i = \text{Encoded}(R_{i-1}) \oplus K_i. \tag{7}$$

In the next step, this block of 48-bit is passed through an S-box to obtain a 32-bit message, i.e., R_i. The S-box split E_i into eight equivalent blocks, where each block requires six inputs and generates four outputs. An XOR operation is carried out between the permuted function and L (L_{i-1}) as shown in (8) and (9), respectively

$$R_i = L_{i-1} \oplus \text{Function}(R_{i-1}, K_i) \tag{8}$$

$$L_i = R_{i-1}. \tag{9}$$

Finally, (8) and (9) are concatenation to form the desired ciphertext.

Once the encrypted ciphertext is generated, the concerned V_i broadcasts it to the nearest S_j. When this message is received by the respective S_j, it converts this message into plaintext using its λ and a 64-bit DES algorithm. The authenticity of the requesting vehicle (V_i) is confirmed via this process as well as λ is shared among trusted modules, i.e., V_i and S_j, in the offline phase, where the probability of intruder infiltration was nearly zero. Because, one of the trusted *MaskIDs* is appended in the ciphertext, so if the respective S_j wants to validate the authenticity of the requesting V_i, then the extracted *MaskID* is searched in the authentic DeviceList (D_L) or ServerList (S_L) as shown in

$$\text{Auth} = \text{MaskID}_i \in \{D_L\} \ \vee \ \text{MaskID}_i \in \{S_L\}. \tag{10}$$

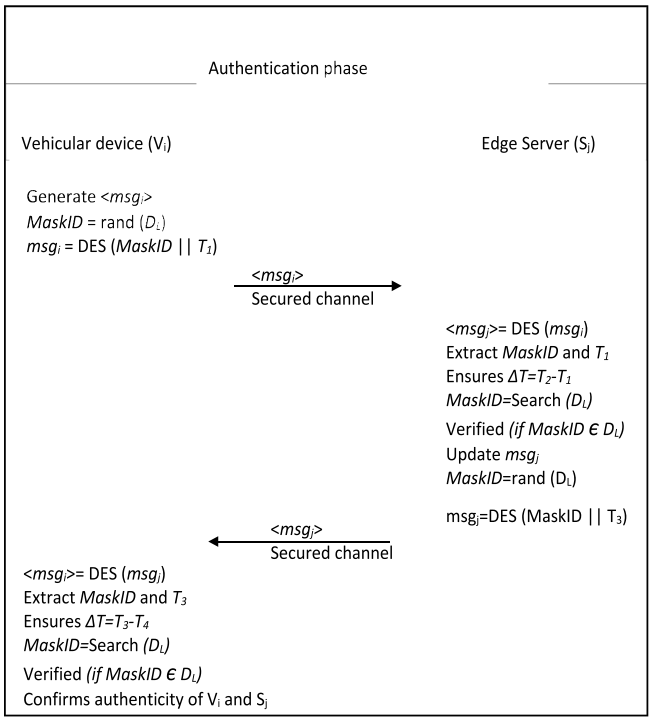


Fig. 3. Graphical representation of authentication for a trusted device.

If *Auth* is *true*, then the requesting device is an authentic device and is allowed to trigger an actual communication session with S_j. A detailed description of the entire authentication process is shown in Fig. 3.

Alternatively, assume that a similar message is being generated by an adversary (A_k) in our DEI framework. To generate this message, the adversary will require the secret key (λ) and a valid *MaskID*. But these crypo primitives were shared among the trusted devices in the offline phase that makes it a cumbersome challenge to crack them. However, if somehow A_k generates and transmits this ciphertext to S_j, then the former will be identified as an adversary due to the following two reasons.

1) Since A_k does not have the required 64-bit λ, the ciphertext message will not be converted into plaintext due to a mismatch with λ, i.e., $\lambda_{A_k} \neq \lambda_{S_j}$. Recall that λ is available to authentic V_i and S_j in the offline phase. However, if we assume that it is an intercepted message that was generated by an authentic V_i, A_k will still not be able to make any changes to it because this message was encrypted with λ that requires 2^{64} attempts to crack it.
2) S_j considers this ciphertext as a suspicious activity because the difference between message generation time (T_G) and message reception time (T_R), i.e., $\triangle T = T_G - T_R$, is greater than the maximum possible level in our DEI framework.

However, if we further assume that the time difference is within the defined bound, then S_j proceeds with normal steps, i.e., conversion into plaintext using its λ and extracting the embedded *MaskID*. This ID is searched within the trusted lists and if a match is encountered then the requesting device is considered as authentic. The respective S_j updates this message

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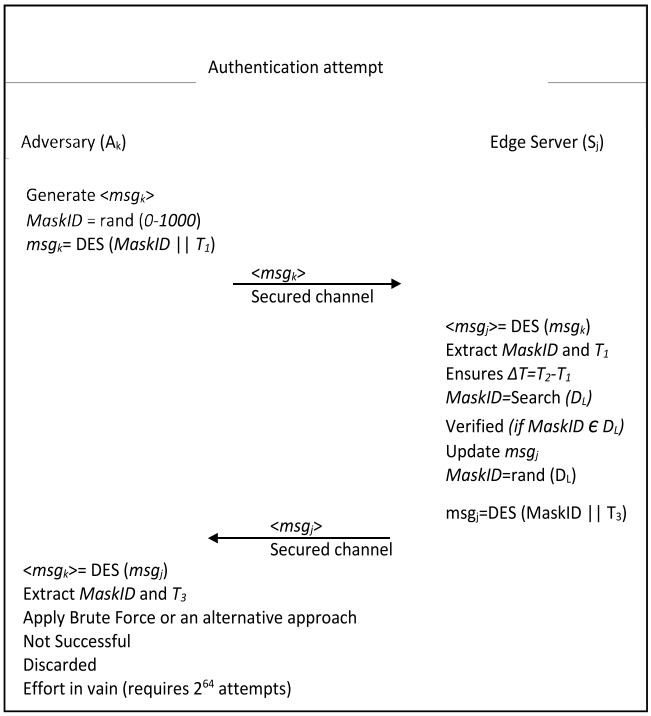


Fig. 4. Graphical representation of authentication attempt by an adversary.

and transmits it back in ciphertext form by encrypting with its λ. Again, if this encrypted message of S_j is intercepted by A_k, the latter will not be able to convert it into plaintext as an authentic λ would be required. Thus, our proposed scheme is pruned against intruders and adversaries attacks for DEI infrastructure. The entire authentication process of both legitimate and adversary devices is graphically shown in Figs. 3 and 4, respectively.

Theorem 1: A vehicular device V_i or an adversary A_k is granted permission to establish a communication session *if and only* if it is an authentic device.

Proof: Let us assume that X and Y represent two distinct classes of devices and edge servers in our IoV-based DEI infrastructure. Consider a vehicle, say $x_i \in X$ requests to initiate a communication session with the intended edge server, say $y_j \in Y$, and at the same time, preserves its privacy as well. For this purpose, x_i sends a request message to y_j, which is encrypted with a 64-bit λ such that <msg> = DES(MaskID || T_1). Generation of this <msg> is only possible if, 1) $x_i \in D_L$, i.e., $MaskID(x_i) \in D_L$ and 2) has a secret 64-bit shared key λ. As x_i is a trusted module it fulfills both conditions. The respective server, say $y_j \in Y$, intercepts the <msg> and tries to decrypt it through the λ. Sever y_j sends permission granted message *iff* DES(<msg>) is successful, ΔT falls within bounds and MaskID $\in D_L$.

Alternatively, if this <msg> is generated by an adversary module A_k, it is identified in the very first step, i.e., <msg> will not be decrypted with the legitimate secret key λ of the server module y_j. Therefore, the respective server, i.e., y_j, either skip it or add it to the adversary modules class. Second, if adversary A_k intercepts an <msg> from a legitimate device, it will not be able to decrypt this message within the defined time interval. If it just forwards this message and intercepts a reply as well, still it will not be able to convert this message into plaintext. However, a legitimate module, i.e., x_i, decrypts this message through its secret key λ and starts communication. Thus, it proves that only legitimate devices are granted permission to initiate communication sessions with the intended server module in the edge-enabled IoT. ■

Theorem 2: An adversary A_k cannot break the proposed model in polynomial time that is $Adv^D_{A_k}(\lambda) \leq ([FR_{1-16}]/[ID+T]) + 128Adv^D_{A_k}ES(\lambda) + ([qs]/[64^{k-1}D])$ where Adv indicates knowledge of the adversary and FR represents Fiestel rounds in the DES. ID and T indicate device ID and timestamp information. qs and D refer to the sent message and secret key length.

Proof: This theorem is proved through four different stages, i.e., SG_0, SG_1, SG_2, and SG_3, respectively, as described below.

SG_0: In this stage, A_k is required to initiate the key cracking process through the intercepted message as shown in (11). Additionally, $Succ^{SG_0}_{A_k}(\lambda)$ means that A_k has captured the message successfully

$$Adv^D_{A_k}(\lambda ij) = \left[64Pr[\mathrm{Succ}^{SG_0}_{A_k}(\lambda ij)]\right]. \tag{11}$$

In SG_1, A_k triggers the execute query to SG_0 to get valuable information, such as the device's ID and λ. However, at SG_1, A_k intercepts ciphertext message which may be of the form $M_1 = (ID_1 \oplus Text \oplus T_1)$, $M_2 = (\mathrm{MaskID}_1 \oplus Text \oplus T_1)$, and $M_3 = (ID_1 \oplus \mathrm{MaskID1} \oplus Text \oplus T_1)$ as per A_k assumption. Then A_k executes the test function, which is specifically designed for the cryptanalysis, to get the secret key $\lambda_{ij} = ID_1 \oplus \mathrm{MaskID} \oplus Text \oplus T_1$ and $ID_2 \oplus \mathrm{MaskID2} \oplus Text \oplus T_2$. However, as λ_{ij} is confidential to V_i and S_j in DEI, it means SG_1 is not different from SG_0 because A_k has not gained any valuable information

$$64Pr\left[Succ^{SG_0}_{A_k}(\lambda_{ij})\right] = 64Pr\left[Succ^{SG_1}_{A_k}(\lambda ij)\right]. \tag{12}$$

In SG_2, A_k adds various other operations, such as perfect forward attack, because it is interested in tampering the original message, but the intercepted message, i.e., msg_i, is encrypted with λ. Additionally, msg_i has MaskID and T_1 values, which are impossible for the A_k to compute. A_k is unable to obtain λ and thus not possible for it to crack DES. Finally, A_k cannot replicate the device's ID and timestamp information in msg_i. Thus, (13) is obtained through birthday paradox and A_k assumption

$$64Pr\left[Succ^{SG_2}_{A_k}(\lambda)\right] - 64Pr\left[Succ^{SG_1}_{A_k}(\lambda)\right] \leq \frac{FR_{1-16}}{64[ID+T]} + 64Adv^{DES}_{A_k}(t). \tag{13}$$

Finally, in SG_3, A_k tries to impersonate a legitimate device and get valuable information, such as ID, MaskID, λ, and timestamp. Furthermore, A_k selects a low entropy λ_k as a secret key to form an encrypted message, i.e., msg_k. Then, we can conclude the success of A_k as shown in

$$Pr\left[Succ^{SG_3}_{A_k}(\lambda)\right] - Pr\left[Succ^{SG_2}_{A_k}(\lambda)\right] \leq \frac{qs}{2^k D}. \tag{14}$$

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TABLE I
EVALUATION OF THE PROPOSED SCHEME IN TERMS OF VARIOUS SECURITY ATTACKS

Security Metric	[21]	[22]	[23]	[24]	[25]	Proposed Scheme
Masquerading Attack (Device)	✓	✓	✓	✓	✓	✓
Anonymity traceability	✓	✓	✓	✓	✓	✓
Masquerading Attack (Device)	✓	✓	✓	✓	✓	✓
Replay Attack	✓	✓	✓	✓	✓	✓
Masquerading Attack (Edge Server)	×	✓	×	×	✓	✓
Eavesdropping Attack	×	×	✓	✓	✓	✓
Brute-Force Attack	✓	✓	✓	✓	✓	✓
Denial of Service Attack	✓	×	✓	✓	✓	✓
Man-in the Middle Attack	✓	×	×	✓	✓	✓
Device Privacy-Preservation	×	×	×	×	×	✓
Server Privacy-Preservation	×	×	×	×	×	✓

Finally, A_k guesses λ through a series of head and trials as depicted in

$$Pr\Big[Succ_{A_k}^{SG_3}(\lambda)\Big] = \frac{1}{64}. \tag{15}$$

By resolving these equations, we obtain

$$\begin{aligned}\frac{Adv_{A_k}^{D}(\lambda)}{64} &= Pr\Big[Succ_{A_k}^{SG_0}(\lambda)\Big] - \frac{1}{64} \\ &= Pr\Big[Succ_{A_k}^{SG_0}(\lambda)\Big] - Pr\Big[Succ_{A_k}^{SG_3}(\lambda)\Big] \\ &= Pr\Big[Succ_{A_k}^{SG_1}(\lambda)\Big] - Pr\Big[Succ_{A_k}^{SG_3}(\lambda)\Big] \\ &\leq \sum_{i=0}^{2}\Big[Pr\Big[Succ_{A_k}^{SG_{i+1}}(\lambda)\Big] - Pr\Big[Succ_{A_k}^{SG_i}(\lambda)\Big]\Big] \\ &= \frac{FR_{1-16}}{64[ID+T]} + 64Adv_{A_k}^{D}ES(\lambda) + \frac{qs}{64^k D}.\end{aligned} \tag{16}$$

Hence, it is proved that

$$Adv_{A_k}^{D}(\lambda) = \leq \frac{FR_{1-16}}{ID+T} + 128Adv_{A_k}^{D}ES(\lambda) + \frac{qs}{64^{k-1}D}. \tag{17}$$

■

IV. INFORMAL SECURITY ANALYSIS

A detailed and thorough performance analysis of our proposed scheme in the context of various attacks is presented here. Furthermore, a comparison against the existing schemes is presented in Table I.

A. Device Impersonation Attack

Consider an adversary A_k that impersonates as an authentic V_i and communicates with a server S_j to initiate a communication session. For this purpose, A_k requires a secret key λ and a *MaskID* to generate an encrypted message. Assume that A_k uses a dummy λ_k and encrypts its generated message, i.e., $<msg_k> = \lambda(A_{ID}||T_I)$, where A_{ID} represents the unique ID of the adversary. The receiving device S_j decrypts this message with its λ_j, i.e., $\lambda_j(<msg>)$. However, $<msg_k>$ was generated with λ_k and it will not be converted into plaintext. Thus, S_j assumes that the message originates from an adversary and refrain the latter from any further communication.

B. Man-in-the-Middle Attack

An A_k may intercept a message $<msg>$, that is generated by a legitimate V_i and tries to modify its content. However, to do so, A_k must have $Predicted_{i=0}^{n}(\lambda_k) = Prob[Succ(\exists_{i=0}^{k}(\lambda)]$ such that $\lambda_k \in legitimate(\lambda)$. However, it is far beyond the operational capacities of these devices, whether legitimate or adversary, to convert $<msg>$ into plaintext. Second, every device has a specified time interval, i.e., $\triangle T$, during which $<msg>$ is expected. Therefore, a delayed message delivery is another metric in identifying the reliability of the $<msg>$. Thus, the proposed authentication approach is pruned against man-in-the-middle attacks, especially in the resource-constrained DEI-enabled IoV.

C. Server Impersonation Attack

Assume that A_k has impersonated to be a legitimate V_i to an S_j. For this purpose, A_k communicates with a legitimate V_i to initiate a communication session. Next, A_k requires λ_j and MaskID of S_j, either to generate an encrypted message or respond to any given query. If we further consider that A_k uses a dummy λ_k, and encrypts its generated message, i.e., $<msg> = \lambda(A_{ID}||T_I)$. Upon reception of $<msg>$, V_i will try to decrypt this message with its λ_i, i.e., $\lambda_i(< msg >)$. However, this $<msg>$ is generated with a nonlegitimate secret key λ_k, where $\lambda_k \notin \lambda_i$ and $\lambda_k \notin \lambda_j$ and, thus, it will not be converted into the respective plaintext. Hence, V_i assumes that the latter is an adversary and avoids any further communication with it.

D. Eavesdropping Attack

In eavesdropping, an A_k tries to intercept every $<msg>$, especially those sent via an unsecured transmission channel by modifying its content or discarding it entirely. In the proposed setup, every message $<msg>$ is encrypted with a secret key λ, a 64-bit DES, and the additional sixteen subkeys for each Fiestel round. Therefore, the initial strategy for A_k would be to use the brute-force attack for extracting λ. However, to do so, A_k must have $Predicted_{i=0}^{n}(\lambda_k) = Prob[Succ(\exists_{i=0}^{k}(\lambda)]$ or $Predicted_{j=0}^{m}(\lambda_k) = Prob[Succ(\exists_{i=0}^{k}(\lambda)]$ such that $\lambda_k \in legitimate(\lambda)$, where m and n are legitimate server and devices, respectively. However, it is far beyond the operational capacities of these devices, whether legitimate or adversaries, to convert $<msg>$ into plaintext. Thus, the proposed approach is pruned against eavesdropping attacks, especially in DEI-enabled IoV infrastructures.

E. Denial of Service Attack

In this attack, A_k aims to make the resources unavailable in DEI-enabled IoV for the legitimate V_i and S_j. For this purpose,

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TABLE II
SIMULATION PARAMETERS OF OUR ML-BASED AUTHENTICATION FOR PRIVACY PRESERVATION IN DISTRIBUTED EDGE-ENABLED IOV

Parameters	Approximate Values
Coverage Region	$800m \times 800m$
Active Devices (V_i)	550 to 1200
Edge Servers (S_j)	5 to 10
Battery Power (E_I)	52000 mAh
Residual Energy (E_r)	Available Power
Average Transmission Cost (P_{T_X})	91.4 mW
Average Receiving Cost (P_{T_R})	59.1 mW
Average Channels Delay	10 ms
Power Consumption in Idle modes	15.4 μW
Power consumption of Transceiver in Idle Modes	1 mW
Average Transmission Range	$500m$
Encryption	64-Bit DES
Topology	Dynamic and Static

A_k applies various techniques, such as excessive transmission of session initiation requests. However, these messages are discarded in the screening process of $<msg>$, because they are generated via a different key, i.e., λ_k, where $\lambda_k \notin \lambda_i$ and $\lambda_k \notin \lambda_j$. Therefore, messages sent by A_k are ignored by V_i and S_j.

V. PERFORMANCE EVALUATION OF THE PROPOSED DISTRIBUTED EDGE INTELLIGENCE SCHEME

In this section, a comparative analysis of the proposed scheme is carried out by comparing it with the existing state-of-the-art schemes in terms of communication, processing, and storage overheads. The experimental results were generated using python and OMNET++. Furthermore, we have considered the onboard batteries of V_i and S_j to be consistent with the battery powers of Wasp motes and Arduino boards. A summarized form of various parameters used in the simulation environment is presented in Table II.

A. Computational Overhead

Our proposed authentication scheme of privacy preservation for DEI uses a 64-bit DES to encrypt every message, which ensures that minimal overhead is incurred in terms of hash functions and XOR operations, as depicted in Table III. Furthermore, computational complexity, i.e., processing and storage, is minimal for our scheme in comparison to the existing schemes. Please note that the computational overhead includes the processing and storage overheads incurred during the offline phase, which is executed preferably once upon the initiation of network operations. Moreover, all the schemes of Table III were thoroughly examined using the average processing time, which is the time when a message $<msg>$ is generated to its encryption. A graphical view of the proposed and existing schemes is shown in Figs. 5 and 6, which confirms the exceptional performance of the proposed authentication scheme. In our experimental results, the expected probability of a potential adversarial attack was kept high.

B. Communication Overhead

In IoV, reliable communication infrastructures are preferred to ensure a successful packet transmission, especially within the defined time interval. Moreover, a minimal bit set transmission leads to congestion-free communication channels with a maximum average packet delivery ratio, an ideal scenario required in resource-constrained networks. For this purpose,

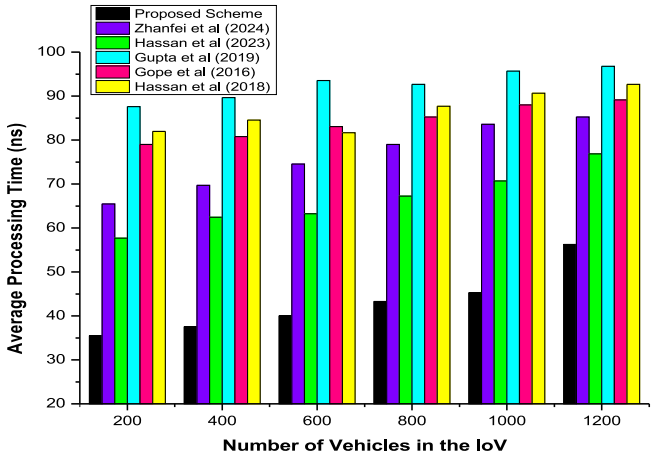


Fig. 5. Average processing time at the vehicular devices.

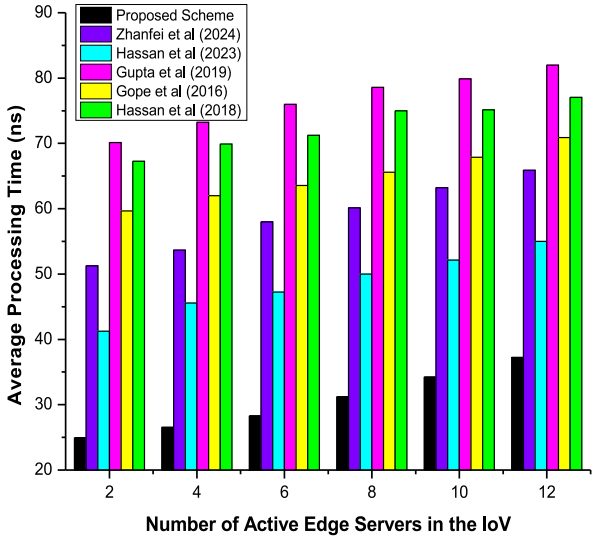


Fig. 6. Average processing time at the edge servers.

the proposed authentication scheme is thoroughly examined for the minimal and maximum bit sets, i.e., 64-bit packet size (data). The proposed ML-based authentication scheme for privacy preservation is compared with the existing schemes in the context of communication overhead in Table IV. The communication overhead of the proposed scheme is 512-bit, including offline phase (256-bit) and online verification phase (256-bit). Additionally, the offline phase is executed only once, as it is required at the beginning only.

Furthermore, a comparative evaluation of these schemes is presented in Fig. 7, where our proposed ML-based authentication scheme for privacy preservation outperforms the existing schemes of a DEI-enabled IoV framework. Please note that these simulations were conducted in realistic scenarios, where the probability of adversarial attacks was kept high.

C. Storage Overhead

In IoV, V_i and S_j rely on the onboard memory or storage module that has a direct proportionality ratio to its smooth functionality. The proposed scheme has ensured that every module, be it the vehicle or edge server, utilizes its memory effectively. In the proposed scheme, each V_i and S_j has to store

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TABLE III
EVALUATION OF THE COMPUTATIONAL OVERHEAD

Algorithms	User/client	Device V_i	Edge Server S_j	Total Cost
Proposed Scheme	-	$2T_h + 3T_{XOR}$	$2T_h + 3T_{XOR}$	$4T_h + 6T_{XOR}$
M. Zhanfei et al. [25]	-	$4T_h + 3T_{ecm}$	$4T_h + 3T_{ecm}$	$8T_h + 6T_{ecm}$
Hassan et al. [26]	$2T_{PA} + 2T_{PM}$	$2T_{E/D} + 2T_{PRNG}$	$3T_{TS} + 1T_{TKNG}$	
Gupta. et al. [24]	$7T_h + 4T_{XOR}$	$4T_h + 4T_{XOR}$	$5T_h + 3T_{XOR}$	$16T_h + 11T_{XOR}$
Gope et al. [27]	$3T_h + T_{XOR}$	—	$9T_h + 4T_{XOR}$	$12T_h + 5T_{XOR}$
Li. et al. [28]	$7T_h + 4T_{XOR}$	$4T_h + 4T_{XOR}$	$5T_h + 3T_{XOR}$	$16T_h + 11T_{XOR}$
Makhalouf. et al. [23]	-	$2T_h + 6T_{XOR}$	$7T_h + 7T_{XOR}$	$9T_h + 13T_{XOR}$
Abdelshafy. et al. [29]	$5T_h + 5T_{XOR}$	$2T_h + 1T_{XOR}$	$2T_h + 6T_{XOR}$	$6t_h + 11T_{XOR}$
Liu. et al. [21]	$3T_h + 3T_{XOR}$	—	$4T_h + 12T_{XOR}$	$7T_h + 19T_{XOR}$
Hasan. et al. [22]	$2T_h + 6T_{XOR}$	$2T_h + 5T_{XOR}$	$3T_h + 3T_{XOR}$	$7T_h + 14T_{XOR}$

TABLE IV
EVALUATION OF COMMUNICATION OVERHEAD

Algorithms	No. of Messages	No. of bits
Proposed Scheme	04	512
Hassan et al. [26]	06	672
M. Zhanfei et al. [25]	3	3,392
Li et al. [21]	04	4,672
Makhalouf. et al. [23]	05	6,144
Gupta. et al. [24]	(05)	3,038
Gope et al. [27]	(04)	3,184
Hasan. et al. [22]	06	32,000
Liu. et al. [21]	(06)	30,620
Abdelshafy. et al. [29]	05	24,546

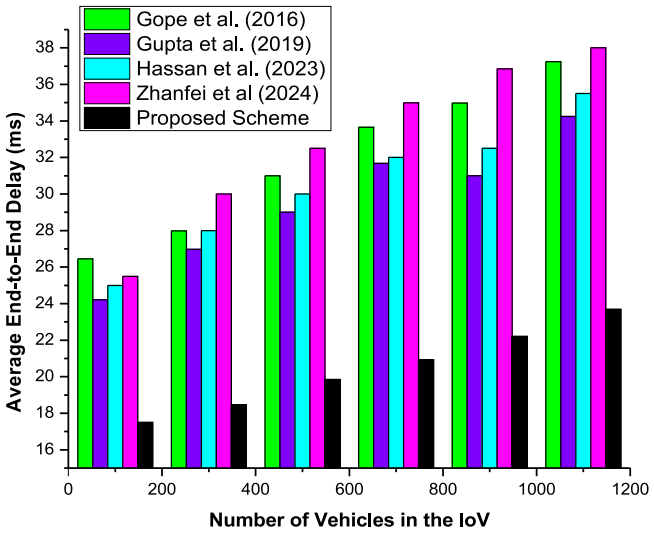


Fig. 7. Average communication overhead in a multihop infrastructure.

TABLE V
COMPARISON OF THE STORAGE OVERHEAD

Algorithms	Modules	Server Gateway
Proposed Scheme	$MasID_{i,j}$ λ	$MasID_i$ λ
M. Zhanfei et al. [25]	$ID_v + T_1 + PK_{RSU}$	$Cert_V + ID_{RSU} + PK_{V_i}$
Gope et al. [27]	–	$x_{id} + Ts_{ug} + \omega + K_{ug} + Ts_{ug_{new}} + Sn_{id_i}^{new} + k_{gs_i}^{new}$
Gupta et al. [24]	$e_j + f_j + x_j + MI_u + MGID_i$	$MSID_j + Z_j + x_i$
Liu et al. [21]	$ID_i + \lambda_i + ID_G + \lambda_G$	n$ID_i + \lambda_i$ m$+ID_G + \lambda_G$
Makhalouf et al. [23]	—	$PI_{RSU} + PK_{TA} + V_i + K_{vi} + ID_{vi}$
Hassan et al. [26]	Q_{AG} (server) + μ_i	Q_{AG}(device) + μ_i

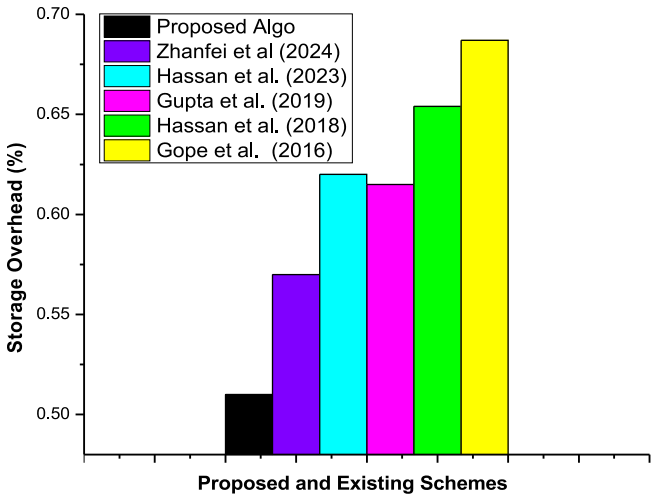


Fig. 8. Average storage overhead.

two lists, i.e., D_L and S_L, along with a 64-bit λ. An extensive analysis of the proposed and existing schemes is presented in Table V. From this table, some of the existing schemes, e.g., [27], has a minimal storage overhead as compared to our scheme. However, the former has a much higher computational and communication overheads, as shown in Figs. 5–7.

Storage overhead is crucial as it has a direct correlation with the smooth functionality of an underlining device. Therefore, we have provided an extensive comparison of the proposed scheme and existing schemes in Table V. In the proposed setup, a module, be it V_i or S_j, needs to keep a list of legitimate devices, i.e., MaskIDs, along with its own λ, which is shared in the offline phase. It is important to note that some of the existing approaches, e.g., [27], either does not use storage or their storage overhead is very low as compared to the proposed scheme. However, these schemes are very expensive, particularly in terms of computational and communication overheads. Further comparison is made for storage overhead in Fig. 8 that reflects the percentage of memory required for storing the crypto primitives of each scheme.

VI. CONCLUSION

In this article, an ML-based authentication scheme was designed to preserve the privacy of vehicular devices and edge servers in a DEI platform. Each entity was required to participate in an offline phase, where two separate lists of

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authentic vehicles and edge servers along with a secret key were shared. Next, each entity, be it the vehicular device or edge server, was trained through a sophisticated ML-based process to classify the devices into authentic and adversary entities. After the successful completion of these two phases, every legitimate entity can differentiate authentic requests from fraudulent ones in a realistic environment of IoV. An extensive informal security analysis was performed to reflect the potential of the proposed scheme against various intruders' attacks. Simulation results have concluded that the proposed scheme is not only pruned against well-known intruder attacks, but it is equally lightweight and effective concerning various performance evaluation metrics, such as computation, communication, and storage overheads.

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